

Embeddings, Rerankers, Small Language Models for Better Search

Trends in 2025

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Embeddings

jina-code-embeddings-1.5b (2025-09-01)

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jina-code-embeddings-0.5b (2025-09-01)

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jina-embeddings-v4 (2025-06-24)

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jina-clip-v2 (2024-11-05)

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jina-embeddings-v3 (2024-09-18)

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jina-colbert-v2 (2024-08-31)

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jina-clip-v1 (2024-06-05)

[HF](#) — [ArXiv](#) — [API](#) — [Blog](#)

jina-colbert-v1-en (2024-02-17)

[HF](#) — [API](#) — [Blog](#)

jina-embeddings-v2-base-es (2024-02-14)

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jina-embeddings-v2-base-code (2024-02-05)

[HF](#) — [API](#) — [Blog](#)

jina-embeddings-v2-base-de (2024-01-15)

[HF](#) — [ArXiv](#) — [API](#) — [Blog](#)

jina-embeddings-v2-base-zh (2024-01-09)

[HF](#) — [ArXiv](#) — [API](#) — [Blog](#)

jina-embeddings-v2-base-en (2023-10-28)

[HF](#) — [ArXiv](#) — [API](#) — [Blog](#)

jina-embedding-b-en-v1 (2023-06-17)

[HF](#) — [ArXiv](#) — [API](#)

Rerankers

jina-reranker-v3 (2025-10-03)

[HF](#) — [ArXiv](#) — [API](#) — [Blog](#)

jina-reranker-m0 (2025-04-08)

[HF](#) — [API](#) — [Blog](#)

jina-reranker-v2-base-multilingual

(2024-06-25)

[HF](#) — [API](#) — [Blog](#)

jina-reranker-v1-turbo-en (2024-04-18)

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jina-reranker-v1-tiny-en (2024-04-18)

[HF](#) — [API](#) — [Blog](#)

jina-reranker-v1-base-en (2024-02-29)

[HF](#) — [API](#) — [Blog](#)

ReaderLMs

ReaderLM-v2 (2025-01-16)

[HF](#) — [ArXiv](#) — [API](#) — [Blog](#)

reader-lm-1.5b (2024-08-11)

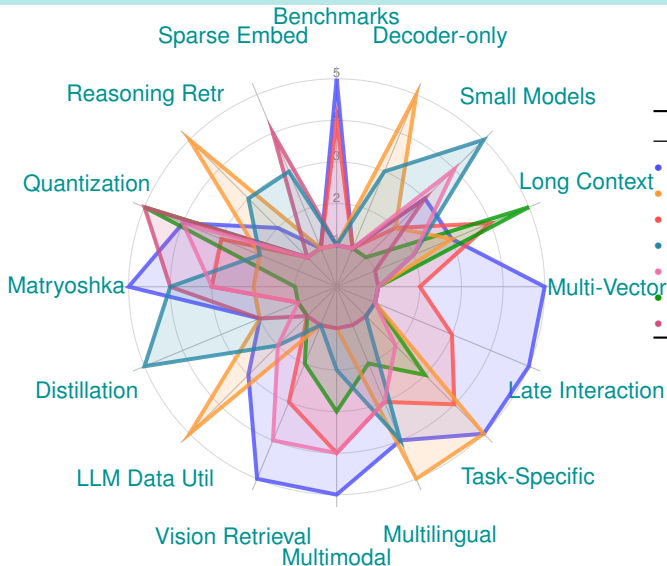
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reader-lm-0.5b (2024-08-11)

[HF](#) — [Blog](#)

Reader [API](#) (2024-04-01) [API](#)

Industry in 2025: Who's Pushing Which Trends



Company	Open-weight?	Avg. Effort
[Redacted]	Open	3.63
[Redacted]	Open	2.63
[Redacted]	Closed	2.50
[Redacted]	Both	2.50
[Redacted]	Open	2.19
[Redacted]	Closed	1.88
[Redacted]	Open	1.81

Trends in 2025

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What Are We Evaluating Beyond BEIR

💡 Trend

BEIR is overfitted. New benchmarks use private test sets, expand to 250+ languages and code domains, enable end-to-end visual document retrieval, and leverage LLM-as-a-judge for evaluation.

Benchmark	Release	Contribution
MMTEB Enevoldsen et al. 2025	Feb 2025	500+ tasks, 250+ languages; instruction & code retrieval
CoIR Li et al. 2024	Jul 2024	10 datasets, 8 code retrieval tasks across 7 domains (ACL 2025)
RTEB Team 2025	Oct 2025	Private test sets prevent overfitting; 20 langs, enterprise domains
ViDoRe v2 Macé et al. 2025	May 2025	Multilingual visual document retrieval benchmark
JinaVDR Günther et al. 2025	Jun 2025	95 tasks, 20 languages for visual documents (tables, charts, diagrams)
AIR-Bench Chen et al. 2024	Dec 2024	LLM-generated heterogeneous eval data across domains

SOTA: MMTEB: 500+ tasks expose multilingual gaps; RTEB private sets reveal 10–15% overfitting drops; AIR-Bench automates diverse domain generation.

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Encoder-only → Decoder-only Architectures

💡 Trend

Decoder-only dominates via foundation LLM backbones from big providers. Encoder-only lacks strong pretrained bases. ModernBERT offers a modern exception with RoPE and Flash Attention.

Model	Contribution
Qwen3 Zhang et al. 2025	Multi-stage pipeline with LLM-synthesized training data
NV-Embed-v2 Lin et al. 2024	Latent attention pooling for task adaptation
Gemini Embedding Lee et al. 2025	3072-dim output for fine-grained semantics
SFR-Embedding-2_R Research 2024	Retrieval-focused training on Mistral-7B base
F2LLM Zhang et al. 2025	SOTA with 6M open-source non-synthetic pairs
Causal2Vec Lin et al. 2025	Contextual token prepending, 85% shorter input
ModernBERT Sounack et al. 2025	Modern encoder-only with RoPE, Flash Attention, 8K context

SOTA: Qwen3-Embedding-8B ranks #1 on MTEB multilingual leaderboard with 70.58 score using multi-stage training with LLM-synthesized data across domains and languages.

Small Models Strike Back

💡 Trend

After NV-Embed pushed sizes to 7B+, the pendulum swings back: Qwen3-0.6B, EmbeddingGemma-300M, and Gemma3-270M prove small models still deliver. Scaling laws reset. Bigger isn't always better.

Model	Params	Contribution
Nomic Embed v1.5 Nussbaum et al. 2024	137M	Long context (8K tokens), outperforms OpenAI embeddings
ModernBERT Sounack et al. 2025	149M/395M	RoPE + Flash Attention, 2× throughput
EmbeddingGemma Vera et al. 2025	300M	Geometric distillation + spread-out regularizer
jina-code-emb Kryvosheieva et al. 2025	494M/1.5B	Domain-specialized SOTA for 15+ programming languages
BGE-M3 Xiao et al. 2024	569M	Unified dense, sparse, and multi-vector in single compact model
jina-emb-v3 Sturua et al. 2024	570M	Task-specific LoRA, outperforms models 12× its size

SOTA: jina-code-embeddings-0.5b achieves 78.41% on code retrieval benchmarks, outperforming Qwen3-Embedding-0.6B by 5 percentage points despite being 20% smaller.

Long Context Embeddings

💡 Trend

512 tokens is BERT-old. Modern embedders and rerankers now support up to 128K tokens, inheriting RoPE, sliding window attention, and other long-context advances from recent LLM base models.

Model	Context	Contribution
Dewey Zhang et al. 2025	128K	Specialized chunk alignment for long documents
jina-emb-v4 Günther et al. 2025	32K	Multimodal architecture with task-specific LoRA
Qwen3 Zhang et al. 2025	32K	Inherited from Qwen3 foundation model
NV-Embed-v2 Lin et al. 2024	32K	Latent attention from Mistral-7B base
GritLM Muennighoff et al. 2024	32K	Unified generation and embedding from Mistral
BGE-M3 Xiao et al. 2024	8K	Multi-granularity retrieval (dense, sparse, colbert)
Nomic v1 Nussbaum et al. 2024	8K	Contrastive pretraining on nomic-bert
jina-emb-v2 Günther et al. 2023	8K	First 8K model to rival ada-002

SOTA: Dewey with 128K achieves state-of-the-art on long-document retrieval via chunk alignment training; jina-emb-v4 with 32K unifies text and image modalities.

Single-Vector vs. Multi-Vector

💡 Trend

Multi-vector trades space and time complexity for better quality. Gaining attention but requires storage backend support and user education on when the tradeoff makes sense.

Model	Contribution
jina-colbert-v2 Jha et al. 2024	Multilingual late interaction with 50% compression
CRISP Veneroso et al. 2025	Clustering-based token pruning for efficiency
WARP Scheerer et al. 2025	3–40× latency reduction via optimized C++ kernels
DocPruner Yan et al. 2025	50–60% storage reduction via adaptive patch pruning
MetaEmbed Xiao et al. 2025	Token-level Matryoshka for test-time adaptation
SOTA: WARP achieves 41× speedup over XTR and 3× over PLAID; DocPruner enables 50–60% storage reduction with negligible quality loss.	

Early vs. Late Interaction

💡 Trend

Exploring when to compute query-document interactions: early, late, or "last-but-not-late." Many interesting ways to attend information in the context window beyond traditional patterns.

Model	Type	Contribution
Late Chunk Xiao et al. 2024	Late	Embed first, then chunk for position-aware retrieval
PyLate Chaffin and Sourty 2025	Late	Flexible training framework for ColBERT-style models
Granite R2 Awasthy et al. 2025	Both	Hybrid bi-encoder and cross-encoder architecture
jina-rerank-v3 Wang et al. 2025	Early	Last-but-not-late interaction with final-token pooling
Cross-encoder Ananthakrishnan et al. 2025	Early	Joint encoding produces reusable embeddings

SOTA: jina-rerank-v3 with 0.6B achieves 61.85 nDCG@10 on BEIR via last-but-not-late interaction with final-token pooling, 10× smaller than generative listwise rerankers.

Architecture Challenges

Open Problems

Leading researchers identify fundamental representation gaps that current architectures cannot address.

- **Implicit semantics gap:** Models capture surface lexical similarity but fail at pragmatic, attitudinal, and sociocultural meaning—no benchmarks evaluate implicit understanding, no architectures beyond contrastive learning address it
- **Vector expressiveness bottleneck:** Single-vector embeddings provably cannot scale to combinatorial query complexity ($d=4096$ breaks at 250M docs)—multi-vector and hybrid architectures mandatory, not optional
- **Reasoning vs. general retrieval specialization:** Models excel at reasoning-intensive OR general retrieval, not both—no unified architecture balances deep semantic understanding with broad applicability

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Task-Specific & Multi-Task Embeddings

💡 Trend

LoRA adapters win over instruction-following for task adaptation. Cheaper to serve, no corpus re-encoding needed. Instructions still useful but LoRA offers better parameter efficiency.

Model	Approach	Contribution
jina-emb-v3 Sturua et al. 2024	LoRA	5 task-specific adapters (retrieval, clustering, classification, similarity, text matching)
jina-emb-v4 Günther et al. 2025	LoRA	3 multimodal adapters (query-doc, semantic similarity, code search)
INSTRUCTOR Su et al. 2022	Instructions	Natural language task descriptions for 330+ diverse tasks
jina-code-emb Kryvosheieva et al. 2025	Instructions	Instruction-following for 15+ programming languages
Qwen3 Zhang et al. 2025	Instructions	Multi-stage training with LLM-synthesized task-specific data

SOTA: Qwen3-Embedding-8B ranks #1 on MTEB multilingual leaderboard with multi-stage training using LLM-synthesized task-specific data; jina-emb-v3 pioneered task-specific LoRA adapters; INSTRUCTOR introduced instruction-following embeddings.

Multilingual Embeddings

💡 Trend

Curse of multilinguality persists: adding more languages degrades per-language performance. LLM-synthesized multilingual data helps but fundamental tradeoff remains. Breadth vs depth still unsolved.

Model	Languages	Contribution
BGE-M3 Xiao et al. 2024	Multilingual	Self-knowledge distillation with multi-granularity
Multilingual E5 Wang et al. 2024	Multilingual	Cross-lingual transfer learning framework
Qwen3 Zhang et al. 2025	Multilingual	LLM-synthesized multilingual training data
Arctic 2.0 Merrick et al. 2024	Multilingual	Matryoshka learning for multilingual embeddings

SOTA: Qwen3-Embedding-8B ranks #1 on MTEB multilingual with 70.58 score via multi-stage training on LLM-synthesized multilingual data.

Multimodal & Omni-Models

💡 Trend

Text+image is table stakes in 2025. True omni-models (text+image+audio+video) emerging but audio/video-only retrieval significantly lags text. Unified semantic spaces across all modalities remain challenging.

Model	Modality	Contribution
GritLM-7B Muennighoff et al. 2024	Text+Image	Unified architecture for generation and embedding
jina-emb-v4 Günther et al. 2025	Text+Image	Task-specific LoRA adapters for diverse retrieval
jina-clip-v2 Koukounas et al. 2024	Text+Image	Multilingual (29 languages) text-image retrieval
jina-rerank-m0	Text+Image	Visual reranking with cross-modal attention
Omni-Embed Xu et al. 2025	Multimodal	NVIDIA unified embedding across modalities
Omni-5 Xiao et al. 2025	Multimodal	Language-centric multimodal representation

SOTA: jina-emb-v4 with 3.8B unifies text and images via single-vector and multi-vector late interaction with task-specific LoRA adapters for query-document, similarity, and code search.

Vision-Based Document Retrieval

💡 Trend

ColPali killed OCR pipelines. End-to-end visual document retrieval from raw images vastly outperforms text-extraction approaches. Layout, tables, charts all understood natively without parsing.

Model	Size	Contribution
ColPali Faysse et al. 2024	0.9B	End-to-end late interaction on visual patches from document images
jina-emb-v4 Günther et al. 2025	3.8B	Unified text and image embeddings with task-specific adapters
ModernVBERT Teiletche et al. 2025	250M	Vision-language encoder with modern architecture
HPC-ColPali Bach 2025	0.9B	Hierarchical patch compression for efficiency

SOTA: ColPali achieves >86% NDCG@10 on slide retrieval via late visual interaction over document image patches, significantly outperforming text extraction-based approaches.

R1-like Reasoning Retrieval Models

💡 Trend

Inspired by DeepSeek R1, reasoning now enters retrieval. Chain-of-thought and RL enhance embeddings and rerankers for complex queries. LREM deployed at Alibaba scale. Test-time reasoning pays off.

Model	Type	Contribution
LREM Tang et al. 2025	Embedding	Reasoning-augmented query embedding with RL; deployed on China's largest e-commerce
Search-R3 Gui and Cheng 2025	Embedding	Reasoning-then-embedding with supervised & RL training
RITE Liu et al. 2025	Embedding	Reasoning-infused embeddings via intermediate reasoning texts
ReasonEmbed Chen et al. 2025	Embedding	38.1 nDCG@10 on BRIGHT benchmark via ReMixer data synthesis
MM-R5 Xu et al. 2025	Reranker	MultiModal reasoning-enhanced reranker via RL
REARANK Zhang et al. 2025	Reranker	Reasoning re-ranking agent via reinforcement learning
Rank-K Yang et al. 2025	Reranker	Test-time reasoning for listwise reranking, 23% over RankZephyr

SOTA: ReasonEmbed achieves record 38.1 nDCG@10 on BRIGHT benchmark; LREM deployed in production at Alibaba; Rank-K achieves 23% improvement via test-time reasoning.

Capabilities Challenges

Open Problems

Recent SOTA models reveal fundamental capability gaps in reasoning, visual understanding, and reliability.

- **Reasoning capacity ceiling:** LLM-based reasoning embeddings (LREM, ReasonEmbed, Search-R3) fundamentally limited by underlying LLM quality—no path to improve reasoning without better LLMs
- **End-to-end visual RAG:** Vision models still require OCR pipelines; pure visual QA + retrieval from image features remains unsolved—“combining visual retrieval and visually grounded query answering” is future work (ColPali)
- **Confidence estimation for reliability:** No reliable method to estimate retrieval confidence for abstention mechanisms—critical for production deployment but absent from current models

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LLM-Based Data Utilization

💡 Trend

Synthetic data everywhere but F2LLM proves curated real data can match SOTA. Quality over quantity. 6M high-quality pairs beat billion-token synthetic dumps. Synthetic helps but curation matters.

Model	Contribution
jina-emb-v4 Günther et al. 2025	VLMs generate multimodal queries and filter low-quality pairs
EmbeddingGemma Lee et al. 2025	LLMs used throughout data preparation: filtering, augmentation, synthesis
Qwen3 Zhang et al. 2025	LLM-synthesized diverse multi-domain query-doc pairs
F2LLM Zhang et al. 2025	High-quality curation of 6M open-source pairs outperforms synthetic data

SOTA: jina-emb-v4 and EmbeddingGemma leverage VLMs/LLMs for end-to-end data preparation; F2LLM achieves SOTA with curated non-synthetic data.

Model Distillation

💡 Trend

Geometric distillation breakthrough: EmbeddingGemma's spread-out regularizer enables 300M to outperform 600M teachers. Multi-stage Matryoshka pipelines beat single-stage. 2025: distillation quality leapt, not just efficiency gains.

Model	Contribution
EmbeddingGemma Vera et al. 2025	Geometric distillation + spread-out regularizer; 300M outperforms 600M
Jasper & Stella Zhang et al. 2024	Multi-stage Matryoshka distillation beats single-stage approaches
LEAF Vujanic and Ruecksties 2025	Teacher-aligned representations via lightweight framework
SAGE Polat et al. 2025	Adaptive guided distillation with dynamic weighting

SOTA: EmbeddingGemma (300M) achieves SOTA among sub-500M via geometric distillation, maintaining lead even when quantized; multi-stage pipelines consistently outperform traditional single-stage distillation.

Matryoshka Representation Learning

💡 Trend

MRL is standard now, not novel. Extensions: token-level Matryoshka (MetaEmbed), 2D variants (layer+dimension). Every new model ships with MRL. Became infrastructure, not innovation.

Model	Contribution
MetaEmbed Xiao et al. 2025	Token-level Matryoshka for multi-vector test-time scaling
jina-code-emb Kryvosheieva et al. 2025	MRL-based code embeddings for 15+ languages
SMEC Zhang et al. 2025	Sequential Matryoshka training for efficiency
Beyond MRL Wen et al. 2025	Sparse coding-based dimensional reduction
2D MRL Wang et al. 2024	Joint layer and dimension Matryoshka training
Jasper Zhang et al. 2024	Multi-stage MRL distillation pipeline

SOTA: MetaEmbed extends MRL to multi-vector retrieval with token-level Matryoshka for test-time adaptation; widely adopted in SOTA models.

Learned Sparse Embeddings

💡 Trend

Sparse scales to LLMs: CSplade at 8B shows sparse works at scale. Extreme sparsity beats dense. Inverted-index efficiency meets neural quality. Underrated comeback for lexical retrieval.

Model	Contribution
CSplade Xu et al. 2025	8B LLM-based sparse retrieval with bidirectional adaptation
Milco Nguyen et al. 2025	Multilingual sparse retrieval outperforming BGE-M3 and Qwen3
BGE-M3 Xiao et al. 2024	Unified dense, sparse, and multi-vector retrieval in single model
ELSER Elastic 2023	Production-scale learned sparse encoder for semantic search
SPLADE Formal et al. 2022	Term expansion with contextualized sparse representations
SOTA: Milco with 30 active dimensions outperforms Qwen3-Embed 0.6B (1024 dims); CSplade scales sparse retrieval to 8B LLMs with competitive performance and smaller indexes.	

Binary Embeddings & Quantization

💡 Trend

1-bit embeddings going mainstream via binary QAT+MRL. Community tools (vLLM, GGUF, MLX) democratize quantized deployment. Storage cost forced innovation. Binary now production requirement, not research.

Model/Tool	Type	Contribution
Cohere binary Reimers and Al 2024	QAT	Binary embeddings trained with contrastive learning
mxbai-v1 Lee and Shakir 2024	QAT	MRL + binary QAT for 64-byte embeddings
BEBR Tencent Gan et al. 2023	QAT	Recurrent binarization for production-scale retrieval
Sentence-T BQ Shakir et al. 2024	PTQ	Post-training binary with rescoring, 96% retention
Scalar int8 Shakir et al. 2024	PTQ	4× compression via calibration dataset
vLLM Snowflake 2025	Serving	High-throughput embedding serving; 16× faster
Unsloth Unsloth 2024	Training	2× faster fine-tuning with LoRA optimizations
GGUF Gerganov 2023	Format	Quantized format for local deployment via llama.cpp
MLX Apple 2023	Format	Array framework for local deployment on Apple Silicon
r/LocalLLaMA r/LocalLLaMA 2024	Community	Hub for sharing quantization and deployment techniques

SOTA: mxbai retains 96% performance at 32× compression via MRL + binary QAT; vLLM delivers 16× throughput gains; community tools democratize quantized deployment.

Optimization Challenges

Open Problems

Training and evaluation infrastructure reveal fundamental bottlenecks in data synthesis, benchmarking, and efficiency.

- **Benchmark coverage gaps:** Models fail outside BEIR/MTEB domains because “benchmarks miss vast swaths of real-world scenarios” (ColPali)—limited benchmark diversity causes poor generalization, not model architecture
- **LLM-dependent synthetic data quality:** All SOTA models rely on LLM-synthesized data, inheriting LLM biases and limitations—“constrained by reasoning capacity of underlying LLMs” limits data quality ceiling
- **Frontier efficiency tradeoff:** Even SOTA models need “smaller, more efficient variants” (jina-v4)—no path to maintain frontier performance while reducing model size below 1B parameters

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Conclusion

Key Trends in 2025

- Decoder-only LLMs dominate embeddings (Qwen3, NV-Embed, F2LLM)
- Small models (<500M) competitive via distillation & reranking
- Long context (32K–128K) & multilingual (100+ languages) standard
- End-to-end visual document retrieval directly from images

Future Directions

- **Small models everywhere:** Sub-500M for local edge deployment, 1–3B for production; both callable as tools by LLMs for retrieval-augmented generation pipelines
- **Omnimodal retrieval:** Unified encoders for text, images, and audio with modality-specific projectors; enables single-cluster search across all data types
- **Staying frontier and open:** Open-source research and models remain competitive with proprietary systems; transparency drives innovation in modern AI

References I

- [1] Kenneth Enevoldsen, Isaac Chung, Imene Kerboua, Márton Kardos, Ashwin Mathur, David Stap, Jay Gala, Wissam Siblini, Dominik Krzemiński, Genta Indra Winata, Saba Sturua, Saiteja Utpala, Mathieu Ciancone, et al. Mmteb: Massive multilingual text embedding benchmark. *International Conference on Learning Representations*, 2025. doi: 10.48550/arXiv.2502.13595.
- [2] Xiangyang Li, Kuicai Dong, Yi Lee, Wei Xia, Yichun Yin, Hao Zhang, Yong Liu, Yasheng Wang, and Ruiming Tang. Coir: A comprehensive benchmark for code information retrieval models. *Annual Meeting of the Association for Computational Linguistics*, 2024. doi: 10.48550/arXiv.2407.02883.
- [3] MTEB Team. Introducing rteb: A new standard for retrieval evaluation. <https://huggingface.co/blog/rteb>, October 2025. Hugging Face Blog.
- [4] Quentin Macé, António Loison, and Manuel Faysse. Vidore benchmark v2: Raising the bar for visual retrieval. In *CoRR*, volume abs/2505.17166, 2025. doi: 10.48550/ARXIV.2505.17166.
- [5] Michael Günther, Saba Sturua, Mohammad Kalim Akram, Isabelle Mohr, Andrei Ungureanu, Bo Wang, Sedigheh Eslami, Scott Martens, Maximilian Werk, Nan Wang, and Han Xiao. jina-embeddings-v4: Universal embeddings for multimodal multilingual retrieval. In *CoRR*, volume abs/2506.18902, 2025. doi: 10.48550/ARXIV.2506.18902.
- [6] Jianlyu Chen, Nan Wang, Chaofan Li, Bo Wang, Shitao Xiao, Han Xiao, Hao Liao, Defu Lian, and Zheng Liu. Air-bench: Automated heterogeneous information retrieval benchmark. *Annual Meeting of the Association for Computational Linguistics*, 2024. doi: 10.48550/arXiv.2412.13102.
- [7] Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. Qwen3 embedding: Advancing text embedding and reranking through foundation models. In *CoRR*, volume abs/2506.05176, 2025. doi: 10.48550/ARXIV.2506.05176.
- [8] Sheng-Chieh Lin, Chankyu Lee, M. Shoeybi, Jimmy Lin, Bryan Catanzaro, and Wei Ping. Mm-embed: Universal multimodal retrieval with multimodal llms. *International Conference on Learning Representations*, 2024. doi: 10.48550/arXiv.2411.02571.
- [9] Jinhyuk Lee, Feiyang Chen, Sahil Dua, Daniel Cer, Madhuri Shanbhogue, Iftekhar Naim, Gustavo Hernández Ábrego, Zhe Li, Kaifeng Chen, Henrique Schechter Vera, Xiaoqi Ren, Shanfeng Zhang, Daniel Salz, Michael Boratko, Jay Han, Blair Chen, Shuo Huang, Vikram Rao, Paul Suganthan, Feng Han, Andreas Douranoglou, Nithi Gupta, Fedor Moiseev, Cathy Yip, Aashi Jain, Simon Baumgartner, Shahrokh Shahi, Frank Palma Gomez, Sandeep Mariserla, Min Choi, Parashar Shah, Sonam Goenka, Ke Chen, Ye Xia, Koert Chen, Sai Meher Karthik Duddu, Yichang Chen, Trevor Walker, Wenlei Zhou, Rakesh Ghiya, Zach Gleicher, Karan Gill, Zhe Dong, Mojtaba Seyedhosseini, Yun-Hsuan Sung, Raphael Hoffmann, and Tom Duerig. Gemini embedding: Generalizable embeddings from gemini. In *CoRR*, volume abs/2503.07891, 2025. doi: 10.48550/ARXIV.2503.07891.
- [10] Salesforce AI Research. Sfr-embedding-2.r: Retrieval-focused embedding model. https://huggingface.co/Salesforce/SFR-Embedding-2_R, 2024. Model card.
- [11] Ziyin Zhang, Zihan Liao, Hang Yu, Peng Di, and Rui Wang. F2llm technical report: Matching sota embedding performance with 6 million open-source data. 2025.
- [12] Ailiang Lin, Zhuoyun Li, and Kotaro Funakoshi. Causal2vec: Improving decoder-only llms as versatile embedding models. In *CoRR*, volume abs/2507.23386, 2025. doi: 10.48550/ARXIV.2507.23386.
- [13] Thomas Sounack, Joshua Davis, Brigitte N. Durieux, Antoine Chaffin, Tom J. Pollard, Eric Lehman, Alistair E. W. Johnson, Matthew McDermott, Tristan Naumann, and C. Lindvall. Bioclinical modernbert: A state-of-the-art long-context encoder for biomedical and clinical nlp. *arXiv.org*, 2025. doi: 10.48550/arXiv.2506.10896.

References II

- [14] Zach Nussbaum, John X. Morris, Brandon Duderstadt, and Andriy Mulyar. Nomic embed: Training a reproducible long context text embedder. *Trans. Mach. Learn. Res.*, 2024. doi: 10.48550/arXiv.2402.01613.
- [15] Henrique Schechter Vera, Sahil Dua, Biao Zhang, Daniel M. Salz, Ryan Mullins, S. Panyam, Sara Smoot, Iftekhar Naim, Joe Zou, Feiyang Chen, Daniel Cer, Alice Lisak, Min Choi, Lucas Gonzalez, Omar Sanseviero, Glenn Cameron, Ian Ballantyne, Kat Black, Kaifeng Chen, Weiyi Wang, Zhe Li, Gus Martins, Jinhyuk Lee, Mark Sherwood, Juyeong Ji, Renjie Wu, Jingxiao Zheng, Jyotinder Singh, Abheesh Sharma, Divya Sreepat, Aashi Jain, Adham Elarabawy, AJ Co, Andreas Dourmanglou, Babak Samari, Ben Hora, B. Potetz, Dahun Kim, Enrique Alfonseca, Fedor Moiseev, Feng Han, Frank Palma Gomez, Gustavo Hernández Abrego, Hesen Zhang, Hui Hui, Jay Han, Karan Gill, Ke Chen, Koert Chen, Madhuri Shanbhogue, Michael Boratko, P. Suganthan, Sai Meher Karthik Duddu, Sandeep Mariserla, Setareh Ariafar, Shanfeng Zhang, Shijie Zhang, Simon Baumgartner, Sonam Goenka, Steve Qiu, T. Dabral, Trevor Walker, Vikram Rao, Waleed Khawaja, Wenlei Zhou, Xiaoqi Ren, Ye Xia, Yichang Chen, Yi ting Chen, Zhe Dong, Zhongli Ding, Francesco Visin, Gael Liu, Jiageng Zhang, Kathleen Kenealy, Michelle Casbon, Ravin Kumar, Thomas Mesnard, Zach Gleicher, C. Brick, Olivier Lacombe, Adam Roberts, Yunhsuan Sung, Raphael Hoffmann, Tris Warkentin, Armand Joulin, Tom Duerig, and Mojtaba Seyedhosseini. Embeddinggemma: Powerful and lightweight text representations. 2025.
- [16] Daria Kryvosheieva, Saba Sturua, Michael Günther, Scott Martens, and Han Xiao. jina-code-embeddings: Code search with matryoshka representation learning. 2025. arXiv:2508.21290.
- [17] Shitao Xiao et al. Bge m3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. 2024. arXiv:2402.03216.
- [18] Saba Sturua, Isabelle Mohr, Mohammad Kalim Akram, Michael Gunther, Bo Wang, Markus Krimmel, Feng Wang, Georgios Mastrapas, Andreas Koukounas, Nan Wang, and Han Xiao. jina-embeddings-v3: Multilingual embeddings with task lora. *arXiv.org*, 2024. doi: 10.48550/arXiv.2409.10173.
- [19] Dun Zhang, Panxiang Zou, and Yudong Zhou. Dewey long context embedding model: A technical report. In *CoRR*, volume abs/2503.20376, 2025. doi: 10.48550/ARXIV.2503.20376.
- [20] Niklas Muennighoff, Hongjin Su, Liang Wang, Nan Yang, Furu Wei, Tao Yu, Amanpreet Singh, and Douwe Kiela. Generative representational instruction tuning. *arXiv.org*, 2024. doi: 10.48550/arXiv.2402.09906.
- [21] Michael Günther, Jackmin Ong, Isabelle Mohr, Alaeddine Abdessaleem, Tanguy Abel, Mohammad Kalim Akram, Susana Guzman, Georgios Mastrapas, Saba Sturua, Bo Wang, Maximilian Werk, Nan Wang, and Han Xiao. Jina embeddings 2: 8192-token general-purpose text embeddings for long documents. In *CoRR*, volume abs/2310.19923, 2023. doi: 10.48550/ARXIV.2310.19923.
- [22] Rohan Jha, Bo Wang, Michael Günther, Saba Sturua, Mohammad Kalim Akram, and Han Xiao. Jina-colbert-v2: A general-purpose multilingual late interaction retriever. In *CoRR*, volume abs/2408.16672, 2024. doi: 10.48550/ARXIV.2408.16672.
- [23] João Veneroso, Rajesh Jayaram, Jinmeng Rao, Gustavo Hernández Ábrego, Majid Hadian, and Daniel Cer. Crisp: Clustering multi-vector representations for denoising and pruning. In *CoRR*, volume abs/2505.11471, 2025. doi: 10.48550/ARXIV.2505.11471.
- [24] Jan Luca Scheerer, Matei Zaharia, Christopher Potts, Gustavo Alonso, and Omar Khatlab. Warp: An efficient engine for multi-vector retrieval. In *SIGIR*, pages 2504–2512, 2025. doi: 10.1145/3726302.3729904.

References III

- [25] Yibo Yan, Guangwei Xu, Xin Zou, Shuliang Liu, James Kwok, and Xuming Hu. Docpruner: A storage-efficient framework for multi-vector visual document retrieval via adaptive patch-level embedding pruning. 2025.
- [26] Zilin Xiao, Qi Ma, Mengting Gu, Chun cheng Jason Chen, Xintao Chen, Vicente Ordonez, and Vijai Mohan. Metaembed: Scaling multimodal retrieval at test-time with flexible late interaction. 2025.
- [27] Han Xiao et al. Late chunking: Contextual chunk embeddings using long-context embedding models. 2024. arXiv:2409.04701.
- [28] Antoine Chaffin and Raphaël Sourty. Pylate: Flexible training and retrieval for late interaction models. 2025. arXiv:2508.03555.
- [29] Parul Awasthy, Aashka Trivedi, Yulong Li, Meet Doshi, R. Bhat, P. Vignesh, Vishwajeet Kumar, Yushu Yang, Bhavani Iyer, Abraham Daniels, Rudra Murthy, Ken Barker, Martin Franz, Madison Lee, Todd A. Ward, S. Roukos, David Cox, L. Lastras, Jaydeep Sen, and Radu Florian. Granite embedding r2 models. *arXiv.org*, 2025. doi: 10.48550/arXiv.2508.21085.
- [30] Feng Wang, Yuqing Li, and Han Xiao. jina-reranker-v3: Last but not late interaction for listwise document reranking. 2025.
- [31] Haritha Ananthakrishnan, Julian Dolby, Harsha Kokel, Horst Samulowitz, and Kavitha Srinivas. Can cross encoders produce useful sentence embeddings? In *CoRR*, volume abs/2502.03552, 2025. doi: 10.48550/ARXIV.2502.03552.
- [32] Hongjin Su, Weijia Shi, Jungo Kasai, Yizhong Wang, Yushi Hu, Mari Ostendorf, Wen tau Yih, Noah A. Smith, Luke Zettlemoyer, and Tao Yu. One embedder, any task: Instruction-finetuned text embeddings. *Annual Meeting of the Association for Computational Linguistics*, 2022. doi: 10.48550/arXiv.2212.09741.
- [33] Liang Wang et al. Multilingual e5 text embeddings: A technical report. 2024. arXiv:2402.05665.
- [34] Luke Merrick et al. Arctic-embed 2.0. In *CoRR*, volume abs/2412.04506, 2024. doi: 10.48550/ARXIV.2412.04506.
- [35] Andreas Koukounas, Georgios Mastrapas, Bo Wang, Mohammad Kalim Akram, Sedigheh Eslami, Michael Günther, Isabelle Mohr, Saba Sturua, Scott Martens, Nan Wang, and Han Xiao. jina-clip-v2: Multilingual multimodal embeddings for text and images. In *CoRR*, volume abs/2412.08802, 2024. doi: 10.48550/ARXIV.2412.08802.
- [36] Mengyao Xu, Wenfei Zhou, Yauhen Babakhin, Gabriel Moreira, Ronay Ak, Radek Osmulski, Bo Liu, Even Oldridge, and Benedikt Schifferer. Omni-embed-nemotron: A unified multimodal retrieval model for text, image, audio, and video. 2025. arXiv:2510.03458.
- [37] Chenghao Xiao, Hou Pong Chan, Hao Zhang, Weiwen Xu, Mahani Aljunied, and Yu Rong. Scaling language-centric omnimodal representation learning. 2025. arXiv:2510.11693.
- [38] Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. Colpali: Efficient document retrieval with vision language models. In *CoRR*, volume abs/2407.01449, 2024. doi: 10.48550/ARXIV.2407.01449.
- [39] Paul Teilletche, Quentin Mac'e, Max Conti, Ant'onio Loison, Gautier Viaud, Pierre Colombo, and Manuel Faysse. Modernvbert: Towards smaller visual document retrievers. 2025.

References IV

- [40] Duong Bach. Hierarchical patch compression for colpali: Efficient multi-vector document retrieval with dynamic pruning and quantization. 2025. arXiv:2506.21601.
- [41] Jianting Tang, Dongshuai Li, Tao Wen, Fuyu Lv, Dan Ou, and Linli Xu. Large reasoning embedding models: Towards next-generation dense retrieval paradigm. 2025.
- [42] Yuntao Gui and James Cheng. Search-r3: Unifying reasoning and embedding generation in large language models. 2025.
- [43] Yuxiang Liu, Tian Wang, Gourab Kundu, Tianyu Cao, Guang Cheng, Zhen Ge, Jianshu Chen, Qingjun Cui, and Trishul M. Chilimbi. Exploring reasoning-infused text embedding with large language models for zero-shot dense retrieval. *arXiv.org*, 2025. doi: 10.48550/arXiv.2509.00276.
- [44] Jianlyu Chen, Junwei Lan, Chaofan Li, Defu Lian, and Zheng Liu. Reasonembed: Enhanced text embeddings for reasoning-intensive document retrieval. 2025.
- [45] Mingjun Xu, Jiahao Liu, Yifan Zhang, Tao Zhang, Chenyang Zhao, Yutao Zhu, and Zhicheng Dou. Mm-r5: Multimodal reasoning-enhanced reranker via reinforcement learning. 2025. arXiv:2506.12364.
- [46] Le Zhang, Bo Wang, Xipeng Qiu, Siva Reddy, and Aishwarya Agrawal. Rearank: Reasoning re-ranking agent via reinforcement learning. 2025. arXiv:2505.20046.
- [47] Eugene Yang, Suraj Nair, David D. Lewis, and Ophir Frieder. Rank-k: Test-time reasoning for listwise reranking. 2025. arXiv:2505.14432.
- [48] Dun Zhang, Jiacheng Li, Ziyang Zeng, and Fulong Wang. Jasper and stella: distillation of sota embedding models. 2024. arXiv:2412.19048.
- [49] Robin Vujanic and Thomas Rueckstiess. Leaf: Knowledge distillation of text embedding models with teacher-aligned representations. 2025. arXiv:2509.12539.
- [50] S. Polat, Poli A. Nemkova, and Mark V. Albert. Synthetic adaptive guided embeddings (sage): A novel knowledge distillation method. 2025. arXiv:2508.14783.
- [51] Biao Zhang, Lixin Chen, Tong Liu, and Bo Zheng. Smec: Rethinking matryoshka representation learning for retrieval embedding compression. 2025. arXiv:2510.12474.
- [52] Tiansheng Wen, Yifei Wang, Zequn Zeng, Zhong Peng, Yudi Su, Xinyang Liu, Bo Chen, Hongwei Liu, Stefanie Jegelka, and Chenyu You. Beyond matryoshka: Revisiting sparse coding for adaptive representation. 2025. arXiv:2503.01776.
- [53] Shuai Wang, Shengyao Zhuang, Bevan Koopman, and Guido Zuccon. 2d matryoshka training for information retrieval. 2024. arXiv:2411.17299.
- [54] Zhichao Xu, Aosong Feng, Yijun Tian 0006, Haibo Ding, and Lin Lee Cheong. Csplade: Learned sparse retrieval with causal language models. In *CoRR*, volume abs/2504.10816, 2025. doi: 10.48550/ARXIV.2504.10816.
- [55] Thong Nguyen, Yibin Lei, Jia-Huei Ju, Eugene Yang, and Andrew Yates. Milco: Learned sparse retrieval across languages via a multilingual connector. 2025.
- [56] Elastic. Elastic learned sparse encoder: Elser.
<https://www.elastic.co/search-labs/blog/elastic-learned-sparse-encoder-elser-retrieval-performance>, 2023. Blog post.
- [57] Thibault Formal, Carlos Lassance, Benjamin Piwowarski, and S. Clinchant. From distillation to hard negative sampling: Making sparse neural ir models more effective. *Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022. doi: 10.1145/3477495.3531857.

References V

- [58] Nils Reimers and Cohere AI. Cohere int8 & binary embeddings - scale your vector database to large datasets. *Cohere Blog*, March 2024. <https://cohere.com/blog/int8-binary-embeddings>.
- [59] Sean Lee and Aamir Shakir. Binary embeddings: 64 bytes per embedding, yee-haw. *mixedbread.ai Blog*, 2024. <https://mixedbread.ai/blog/binary-mrl>.
- [60] Yukang Gan, Yixiao Ge, Chang Zhou, Shupeng Su, Zhouchuan Xu, Xuyuan Xu, Quanchao Hui, Xiang Chen, Yexin Wang, and Ying Shan. Binary embedding-based retrieval at tencent. In *KDD*, pages 4056–4067, 2023. doi: 10.1145/3580305.3599782.
- [61] Aamir Shakir, Tom Aarsen, and Sean Lee. Binary and scalar embedding quantization for significantly faster & cheaper retrieval. *Hugging Face Blog*, 2024. <https://huggingface.co/blog/embedding-quantization>.
- [62] Snowflake. Scaling vllm for embeddings: 16x throughput and cost reduction. <https://www.snowflake.com/en/engineering-blog/embedding-inference-arctic-16x-faster>, May 2025. Snowflake Engineering Blog.
- [63] Unsloth. Unsloth: 2x faster fine-tuning with 80% less memory. <https://github.com/unslothai/unsloth>, 2024. GitHub repository.
- [64] Georgi Gerganov. Gguf: Gpt-generated unified format for llama.cpp. <https://github.com/ggerganov/llama.cpp>, 2023. llama.cpp format specification.
- [65] Apple. Mlx: An array framework for apple silicon. <https://github.com/ml-explore/mlx>, 2023. Apple MLX Framework.
- [66] r/LocalLLaMA. r/localllama: Community-driven llm development and deployment. <https://www.reddit.com/r/LocalLLaMA/>, 2024. Reddit community.